Learning Objective 1.3: I can determine the polarization of an antenna and describe the bandwidth characteristics of common antenna types.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Identify the polarization.** For each plane wave propagating in $+\hat{z}$, identify the polarization (linear, RHCP, LHCP, or elliptical) and — where linear — the tilt angle ψ measured from $\hat{x}$.

   (a) $\vec{E} = \hat{x}3\cos(\omega t - kz)$

   Only an $\hat{x}$ component — **linear along $\hat{x}$** ($\psi = 0°$).

   (b) $\vec{E} = \hat{x}2\cos(\omega t - kz) + \hat{y}2\cos(\omega t - kz)$

   Equal amplitudes, in phase — **linear, tilted** $\psi = +45°$.

   (c) $\vec{E} = \hat{x}5\cos(\omega t - kz) + \hat{y}5\cos(\omega t - kz - 90°)$

   Equal amplitudes, $\delta = -90°$, $+\hat{z}$ propagation — **RHCP**.

   (d) $\vec{E} = \hat{x}5\cos(\omega t - kz) + \hat{y}5\cos(\omega t - kz + 90°)$

   $\delta = +90°$ — **LHCP**.

   (e) $\vec{E} = \hat{x}3\cos(\omega t - kz) + \hat{y}1\cos(\omega t - kz - 90°)$

   Unequal amplitudes, $\delta = -90°$ — **right-hand elliptical**, AR $= 3$ (≈ 9.5 dB).

2. **Axial ratio.** A satellite downlink antenna is spec'd circularly polarized with AR $\leqslant 3$ dB across its operating band.

   (a) Convert the 3 dB axial ratio to a linear ratio (major/minor axis).

   AR = $\boxed{\approx 1.41 \text{ (unitless)}}$

   $$\text{AR} = 10^{3/20} \approx 1.41 \quad (\text{i.e. } \sqrt{2})$$

   (b) Write the two orthogonal linear components (up to a common scale) with the appropriate 90° phase, assuming the ellipse major axis is along $\hat{\mathbf{x}}$.

   $$E_x(t) = 1 \cdot \cos(\omega t), \qquad E_y(t) = 0.707 \cos(\omega t \pm 90^\circ)$$

   Sign of the phase sets the RHCP vs LHCP sense.

   (c) An $\hat{\mathbf{x}}$ receiver picks off E_x; a $\hat{\mathbf{y}}$ receiver picks off E_y. Compute the ratio of received *power* in dB. Should it match part (a)?

   P_x/P_y = $\boxed{\approx 3.0 \text{ dB}}$

   $$\frac{P_x}{P_y} = \left(\frac{1.0}{0.707}\right)^2 = 2.0 \longrightarrow 10\log_{10}(2.0) \approx 3.0 \text{ dB}$$

   Yes — the axial ratio in dB equals the received-power difference between the two aligned linear receivers.

3. **Polarization loss factor.**

(a) A ground station transmits a *linearly* polarized signal along $\hat{\mathbf{x}}$. A satellite receiver is **RHCP** and pointed correctly. Compute the PLF in dB.

PLF = $\boxed{0.5 = -3.0 \text{ dB}}$

Linear ↔ CP splits the incident power evenly between the two senses:

$$\text{PLF} = 0.5 \longrightarrow 10\log_{10}(0.5) = -3.0 \text{ dB}$$

(b) Same ground station, but the satellite receiver is **LHCP**. Compute the PLF.

PLF = $\boxed{0.5 = -3.0 \text{ dB}}$

Still -3.0 dB . A linear wave contains equal RHCP and LHCP content, so linear-to-CP coupling does not depend on the sense of the CP antenna.

(c) Two linear whip antennas: radio A vertical, radio B tilted $\psi = 30^\circ$ from vertical. Compute the PLF.

PLF = $\boxed{0.75 = -1.2 \text{ dB}}$

$$\begin{aligned} \text{PLF} &= \cos^2\psi = (0.866)^2 \approx 0.75 \\ &\longrightarrow 10\log_{10}(0.75) \approx -1.2 \text{ dB} \end{aligned}$$

(d) In one sentence, explain why aircraft-to-ground links almost always specify circular polarization on at least one end.

Airframe attitude (roll/pitch/yaw) sweeps a linear aircraft antenna through every tilt angle ψ relative to the ground station, which for two linear antennas means an occasional deep null; making one end CP holds the loss near 3 dB regardless of attitude.

4. **Impedance bandwidth from a VSWR spec.** An antenna is spec'd "VSWR $\leqslant 2$ from 2.30GHz to 2.55GHz."

(a) What is the impedance bandwidth in MHz?

$$\text{BW} = 2.55\text{GHz} - 2.30\text{GHz} = 250\text{MHz}$$

BW = $\boxed{250\text{MHz}}$

(b) What is the fractional bandwidth (FBW) in percent?

$$f_c = \frac{2.30 + 2.55}{2} = 2.425\text{GHz}$$
$$\text{FBW} = \frac{250}{2425} \approx 10.3\%$$

FBW = $\boxed{\approx 10.3\%}$

(c) Convert VSWR $= 2$ to $|\Gamma|$ and to return loss in dB.

$$|\Gamma| = \frac{2-1}{2+1} = \frac{1}{3} \approx 0.333$$
$$\text{RL} = -20\log_{10}(0.333) \approx 9.5\text{ dB}$$

$|\Gamma|$ = $\boxed{\approx 0.333 \text{ (unitless)}}$

RL = $\boxed{\approx 9.5\text{ dB}}$

(d) Narrowband, broadband, or ultra-wideband? Name the likely antenna family, using the bandwidth table from Lesson 3.

FBW $\approx 10\%$ is **broadband**. By the L3 table the natural fit is a **dipole ($\lambda/2$)**, quoted at 8–15%; a **slot** at 5–10% is defensible if the antenna has to live in a skin or a wall. A patch (1–5%) would need matching tricks to stretch this far, and a log-periodic (10:1 RBW) is overkill for a 10% requirement.

5. **Fractional and ratio bandwidth.**

(a) A cellular antenna covers 698–960MHz (low) and 1.71–2.69GHz (high). Compute the FBW and RBW of each band.

low = $31.6\%,\ 1.38{:}1$

high = $44.5\%,\ 1.57{:}1$

Low: $f_c = 829\text{MHz}$

$$\text{FBW} = \frac{262}{829} \approx 31.6\%$$

$$\text{RBW} = \frac{960}{698} \approx 1.38{:}1$$

High: $f_c = 2200\text{MHz}$

$$\text{FBW} = \frac{980}{2200} \approx 44.5\%$$

$$\text{RBW} = \frac{2690}{1710} \approx 1.57{:}1$$

(b) A UWB radar antenna covers 3.1–10.6GHz. Compute its RBW. Does it meet the FCC UWB threshold of RBW $\geqslant 2{:}1$?

RBW = $\approx 3.42{:}1$

$$\text{RBW} = \frac{10.6}{3.1} \approx 3.42{:}1$$

Yes — well above the $2{:}1$ threshold.

(c) Which antenna family would you pick for each of the three bands? Give a one-sentence reason for each that names the *mechanism* buying the bandwidth — resonant, traveling-wave, or self-scaling.

Low ($\approx 32\%$): a broadband monopole or dipole, fattened or sleeved. It is still a *resonant* radiator, but a thick element has a low Q, and a low Q is exactly how a resonant structure buys tens of percent. **High ($\approx 45\%$):** a slot or a dipole again — resonant, but at these frequencies the element is physically small enough that you can afford to make it electrically fat, which is the same low-Q mechanism pushed further. **UWB ($3.4{:}1$):** a Vivaldi (tapered slot) or a log-periodic. These are *traveling-wave* and *self-scaling* structures: a different part of the antenna does the radiating at each frequency, so there is no single resonance to run out of and no matching network to fight. (A PIFA at the low band or a stacked patch at the high band are legitimate extras — but justify them by the same Q argument, not by name.)

6. **Chu-Harrington intuition.** A design must fit inside a sphere of radius $a = 15$ mm and operate at $f = 915$MHz (ISM band).

   (a) Compute λ, k, and the electrical size ka.

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{915 \times 10^6} \approx 0.328\text{m}$$
$$k = \frac{2\pi}{\lambda} \approx 19.2 \text{ rad/m}$$
$$ka = 19.2(0.015) \approx 0.288$$

λ = $\boxed{\approx 0.328\text{m}}$

k = $\boxed{\approx 19.2 \text{ rad/m}}$

ka = $\boxed{\approx 0.288 \text{ (unitless)}}$

   (b) Estimate the minimum Q from $Q_{\text{min}} \approx \frac{1}{(ka)^3} + \frac{1}{ka}$.

$$Q_{\text{min}} \approx \frac{1}{0.0239} + 3.47 \approx 41.8 + 3.5 \approx 45$$

Q_{min} = $\boxed{\approx 45 \text{ (unitless)}}$

   (c) Estimate the maximum fractional bandwidth $\text{FBW}_{\text{max}} \approx 1/Q_{\text{min}}$ (for VSWR $\leqslant 2$).

$$\text{FBW}_{\text{max}} \approx \frac{1}{45} \approx 2.2\% \approx 20\text{MHz at } 915\text{MHz}$$

FBW_{max} = $\boxed{\approx 2.2\% \text{ (20MHz)}}$

   (d) The designer wants 100 MHz of usable bandwidth at 915 MHz. Possible in a 15 mm sphere?

   No — 100MHz is FBW $\approx 11\%$, roughly $5\times$ the Chu-Harrington ceiling for a 15 mm sphere. It requires a larger antenna, or resistive loading / matching-network absorption that trades gain for bandwidth.

7. **Reading the datasheet.** A vendor offers a GPS patch specified as **RHCP**, AR $\leqslant 3$ dB, "bandwidth 4% (VSWR $\leqslant 2$)". You are choosing the antenna for the other end of the link.

   (a) The datasheet defines handedness *from the receiver looking back toward the incoming wave*. Which convention is that, what is the part's sense in **IEEE** terms, and what goes wrong if you pair it with a satellite transmitting IEEE RHCP?

   That is the **optics/physics** convention, opposite to IEEE. The part is IEEE **LHCP**. Paired with an IEEE RHCP satellite it is a *sense mismatch* — ideally a null, and in practice only the few dB of rejection set by the real axial ratio (part (e)). This is a link-killer, not a few-dB penalty.

   (b) Convert AR $= 3$ dB to a linear ratio A, and give A^2.

   $$A = 10^{3/20} \approx 1.413, \qquad A^2 \approx 1.995 \approx 2.0$$

   A = $\boxed{1.413,\ A^2 = 1.995}$

   (c) A *linear* antenna receives from this patch. Using $\mathrm{PLF}(\psi) = \dfrac{A^2\cos^2\psi + \sin^2\psi}{A^2+1}$, find the best and worst PLF over all orientations ψ, in dB.

   best = $\boxed{\approx -1.8 \text{ dB}}$

   worst = $\boxed{\approx -4.8 \text{ dB}}$

   $$p_{\text{max}} = \frac{A^2}{A^2+1} \approx \frac{2}{3} \longrightarrow 10\log_{10}(0.667) \approx -1.8 \text{ dB}$$

   $$p_{\text{min}} = \frac{1}{A^2+1} \approx \frac{1}{3} \longrightarrow 10\log_{10}(0.333) \approx -4.8 \text{ dB}$$

   Not the flat -3 dB of the ideal table — and you rarely control ψ.

   (d) Show that the peak-to-null swing in dB equals the axial ratio in dB.

   $$\frac{p_{\text{max}}}{p_{\text{min}}} = \frac{A^2/(A^2+1)}{1/(A^2+1)} = A^2$$

   $$10\log_{10}\left(A^2\right) = 20\log_{10}(A) = \text{AR}_{\text{dB}}$$

   Here $10\log_{10}(1.995) \approx 3.0$ dB, matching the spec. The result is exact, not a rule of thumb.

(e) The far end is an opposite-sense CP antenna, also AR $= 3$ dB, major axes aligned. Worst-case rejection is $\left(\frac{A^2-1}{A^2+1}\right)^2$. Evaluate it and compare with the idealized PLF table.

rejection = $\boxed{\approx -9.6 \text{ dB}}$

Carry the exact $A = 10^{3/20}$, so $A^2 = 1.995$:

$$\frac{A^2-1}{A^2+1} = \frac{0.995}{2.995} = 0.3322$$
$$(0.3322)^2 = 0.1104$$
$$10\log_{10}(0.1104) \approx -9.6 \text{ dB}$$

(The $A^2 = 2$ shortcut gives $1/9 = 0.111 \rightarrow -9.5$ dB; the course number is -9.6 dB.) The ideal table says $-\infty$ dB. Real hardware gives roughly 10 dB of cross-pol isolation — useful, but nowhere near a null.

(f) Name two things you must ask before believing the antenna works over the advertised 4%.

Which bandwidth — 4% is the *impedance* BW at VSWR $\leqslant 2$, while the axial-ratio and pattern bandwidths are narrower and go unquoted — and **over what scan angle**, since AR is normally specified at boresight only and degrades off-axis. A defensible third: at what VSWR threshold, since the number moves with the bar.

Documentation: ____